

DUNG BUI

Postdoctoral Researcher in Cryptography · Northwestern University

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ABOUT

My research focuses on cryptography, including secure multiparty computation, zero-knowledge proofs, and post-quantum cryptography. I am particularly interested in bridging theory and practice: designing protocols that are both provably secure and efficient enough for real-world use.

POSITIONS

Northwestern University — Postdoctoral Researcher 2026 – now
Department of Computer Science, Evanston, IL, USA.
Host: Xiao Wang.

Sorbonne Université, CNRS, LIP6 — Postdoctoral Researcher 2025 – 2026
Paris, France.
Hosts: Eleni Diamanti, Alex B. Grilo, Ludovic Perret.

EDUCATION

PhD in Computer Science — Université Paris Cité, CNRS, IRIF 2021 – 2025
Thesis: *Efficient Secure Computation from Correlated Pseudorandomness*.
Advisor: Geoffroy Couteau.
Honorable mention, PhD Thesis Prize of GDR Sécurité (CNRS).

MSc in Mathematics, Cryptology, Coding & Applications (CRYPTIS) — Université de Limoges 2019 – 2021
Highest honours, ranked 1st.
Fully funded by the Vingroup Scholarship Program.

BSc in Pure Mathematics — École Normale Supérieure de Hanoi 2015 – 2019
Hanoi, Vietnam. Highest honours, top 5%.

RESEARCH EXPERIENCE

NTT Social Informatics Laboratories — Visiting Researcher Jun – Aug 2024
Tokyo, Japan.
Host: Masayuki Abe.
Multi-round zero-knowledge proofs and applications to advanced post-quantum signatures based on MPC.

COSIC, KU Leuven — Visiting Researcher Jul – Aug 2023
Leuven, Belgium.
Host: Nigel Smart.
Efficient post-quantum signatures from multiparty computation.

IRIF, Université Paris Cité — Research Intern Mar – Aug 2021
Paris, France.
Advisor: Geoffroy Couteau.
Efficient MPC protocols from pseudorandom correlation generators.

XLIM, Université de Limoges — Research Intern

Jun – Aug 2020

Limoges, France.

Advisor: Duong Hieu Phan.

Private set intersection and its applications.

PUBLICATIONS

Author lists follow the alphabetical convention of cryptography. Full versions on DBLP.

2026► **Critical Round in Multi-Round Proofs: Compositions and Transformation to Trapdoor Commitments**M. Abe, D. Balbás, **D. Bui**, M. Ohkubo, Z. Shang, A. Takahashi, M. Tibouchi*EUROCRYPT 2026 — Annual Int. Conference on the Theory and Applications of Cryptographic Techniques***2025**► **Efficient Fuzzy Labeled PSI from Vector Ring-OLE****D. Bui**, K. Cong*CANS 2025 — International Conference on Cryptology and Network Security* · **Best Paper Award**► **New Framework for Structure-Aware PSI From Distributed Function Secret Sharing****D. Bui**, G. Garimella, P. Miao, P.V.L. Pham*ASIACRYPT 2025 — Int. Conference on the Theory and Application of Cryptology and Information Security*► **Structured-Seed Local Pseudorandom Generators and their Applications**B. Applebaum, **D. Bui**, G. Couteau, N. Melissaris*RANDOM 2025 — International Conference on Randomization and Computation*► **Improved All-but-One Vector Commitment with Applications to Post-Quantum Signatures****D. Bui**, K. Cong, C. Delpech de Saint Guilhem*ACISP 2025 — Australasian Conference on Information Security and Privacy*► **Efficient Multi-instance Vector Commitment and Application to Post-quantum Signatures****D. Bui***ACISP 2025 — Australasian Conference on Information Security and Privacy***2024**► **Faster Signatures from MPC-in-the-Head****D. Bui**, E. Carozza, G. Couteau, D. Goudarzi, A. Joux*ASIACRYPT 2024 — Int. Conference on the Theory and Application of Cryptology and Information Security*► **FOLEAGE: $\mathbb{F}_4$ OLE-Based Multi-Party Computation for Boolean Circuits**M. Bombar, **D. Bui**, G. Couteau, A. Couvreur, C. Ducros, S. Servan-Schreiber*ASIACRYPT 2024 — Int. Conference on the Theory and Application of Cryptology and Information Security*► **Fast Public-Key Silent OT and More from Constrained Naor-Reingold****D. Bui**, G. Couteau, P. Meyer, A. Passelègue, M. Riahinia*EUROCRYPT 2024 — Annual Int. Conference on the Theory and Applications of Cryptographic Techniques***2023**► **An Efficient ZK Compiler from SIMD Circuits to General Circuits****D. Bui**, H. Chu, G. Couteau, X. Wang, C. Weng, K. Yang, Y. Yu*Journal of Cryptology, 2024*► **Improved Private Set Intersection for Sets with Small Entries****D. Bui**, G. Couteau*PKC 2023 — IACR International Conference on Practice and Theory of Public-Key Cryptography***TALKS**

Critical Round in Multi-Round Zero-Knowledge Proofs

- Vietnam Institute for Advanced Study in Mathematics (VIASM), May 2026 (virtual)
- ALMASTY seminar, Sorbonne Université, April 2026
- Cybersécurité DVRC/ESILV workshop, Paris, March 2026
- INRIA – C2 (Codage & Cryptographie) seminar, Paris, January 2026

Efficient Fuzzy Labeled PSI from Vector Ring-OLE

- CANS 2025, Osaka, November 2025

Optimized MPC-in-the-Head Based Signatures from Puncturable PRF

- Crypto Seminar, NTT Social Informatics Laboratories, Tokyo, November 2025
- Vietnam Institute for Advanced Study in Mathematics (VIASM), April 2025 (virtual)
- Crypto Student Seminar, CWI Amsterdam, December 2024
- Crypto Day, Télécom Paris (Institut Polytechnique de Paris), November 2024

FOLEAGE: $\mathbb{F}_4$ OLE-Based Multi-Party Computation for Boolean Circuits

- Crypto Café, Florida Atlantic University, September 2025 (virtual)
- ALMASTY seminar, Sorbonne Université, March 2025

Improved All-but-One Vector Commitment with Applications to Post-Quantum Signatures

- ACISP 2025, Wollongong, July 2025

Faster Signatures from MPC-in-the-Head

- ASIACRYPT 2024, Kolkata, December 2024

Efficient MPC from Correlated Randomness

- Crypto Seminar, NTT Social Informatics Laboratories, Tokyo, June 2024

Fast Public-Key Silent OT and More from Constrained Naor-Reingold

- AlgoCRYPT seminar, J.P. Morgan, New York, May 2024 (virtual)
- Algorithms and Complexity seminar, IRIF, Paris, May 2024

Efficient PSI from Pseudorandom Correlation Generators

- Crypto Seminar, COSIC (KU Leuven), July 2023
- Vietnam Institute for Advanced Study in Mathematics (VIASM), May 2023 (virtual)
- Algorithms and Complexity seminar, IRIF, Paris, April 2023
- Crypto Student Seminar, CWI Amsterdam, July 2022 (virtual)

Improved Private Set Intersection with Small Entries

- PKC 2023, Atlanta, May 2023

Private Set Intersection from Correlated Randomness

- Journées C2 (Codage & Cryptographie), Hendaye, April 2022

FELLOWSHIPS & AWARDS**Honorable Mention, PhD Thesis Prize, GDR Sécurité (CNRS)**

2025

Annual national award for an excellent PhD thesis in information security in France.

Best Paper Award, CANS

2025

For *Efficient Fuzzy Labeled PSI from Vector Ring-OLE*.**Fully-Funded Doctoral Grant, DIM Math Innov – FSMP**

2021

Awarded by the Île-de-France Region; up to 9 PhD fellowships across mathematics and computer science.

Silver Medal, NSUCRYPTO — International Olympiad in Cryptography	2021
Fully-Funded Scholarship, Vingroup Scholarship Program For the CRYPTIS Master's program at the Université de Limoges.	2019
VIASM Scholarship, National Program for the Development of Mathematics Vietnam Institute for Advanced Study in Mathematics.	2017 – 2019
Bronze Medal, Vietnam National Mathematical Olympiad (VMO) Part of the selection process for the International Mathematical Olympiad (IMO).	2015

PROFESSIONAL SERVICE

Program Committee

APKC 2025, LATINCRYPT 2025, APKC 2026, CCS 2026.

External Reviewer

CRYPTO 2025 ($\times 3$), 2026 ($\times 3$); EUROCRYPT 2025, 2026 ($\times 2$); ASIACRYPT 2025; PKC 2026; ICALP 2026; CCS 2025; ACNS 2025; PROVSEC 2025; IEEE TIFS 2023; CSF 2022–2023; TCC 2022.

Organizer, IRIF Crypto Seminar	2022 – 2025
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OUTREACH

Member, Advanced Cryptography Research Lab	2025 – now
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Joint initiative between VIASM and the Vietnam Government Information Security Commission (acr-lab.github.io).

Featured interview, “Meet the Women Experts in Quantum Science”	2025
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QICS, Sorbonne Université — on career paths and motivation in post-quantum cryptography.

Co-organizer, educational board game on quantum mechanics (QATS)	2025
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Led activities during “Quantum Day (Lumière sur la quantique)”, organized by DIM QuanTiP, PCQT, Quantum Saclay, Atouts Sciences, QICS, and la SFP.

Featured interview, <i>Tuổi Trẻ</i> — Vietnamese national magazine	2019
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On career paths and motivation in cryptography.

Founder & author, Vcryptis Blog	2019 – 2021
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Practical advice on student life and academics at the Université de Limoges (vcryptis.tech.blog).

LANGUAGES

Vietnamese (native) · English (professional proficiency) · French (DEL F B2).

REFERENCES

Geoffroy Couteau — CNRS, IRIF, Université Paris Cité

PhD advisor · couteau@irif.fr

Xiao Wang — Northwestern University

Postdoctoral host · wangxiao@northwestern.edu

Masayuki Abe — NTT Social Informatics Laboratories

Research collaborator and host · msyk.abe@ntt.com